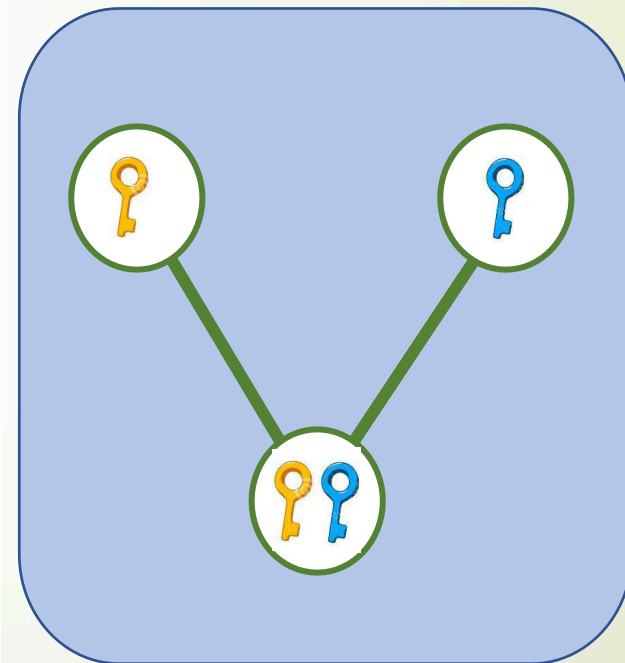
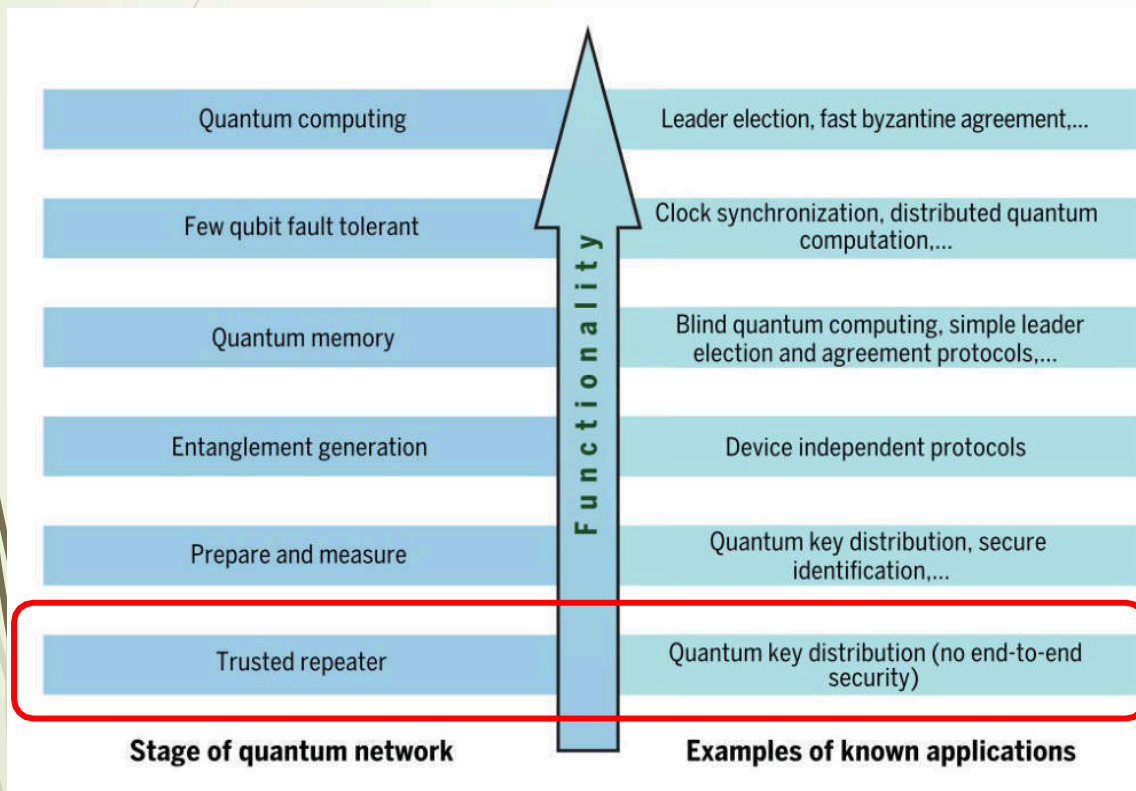
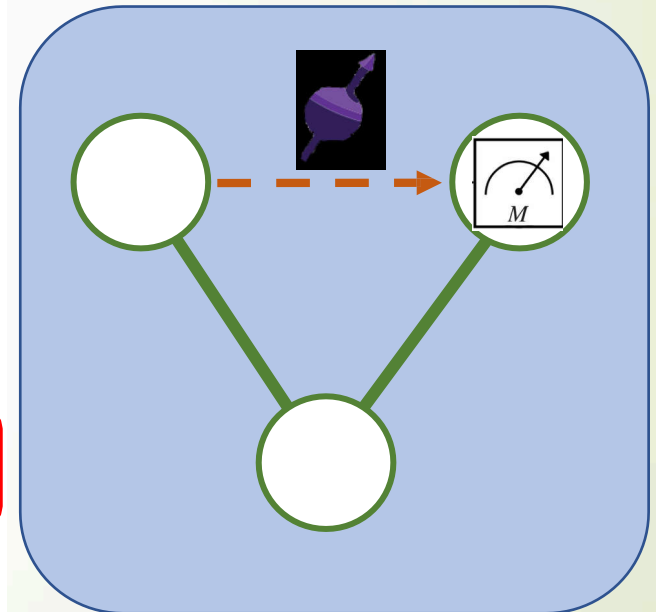
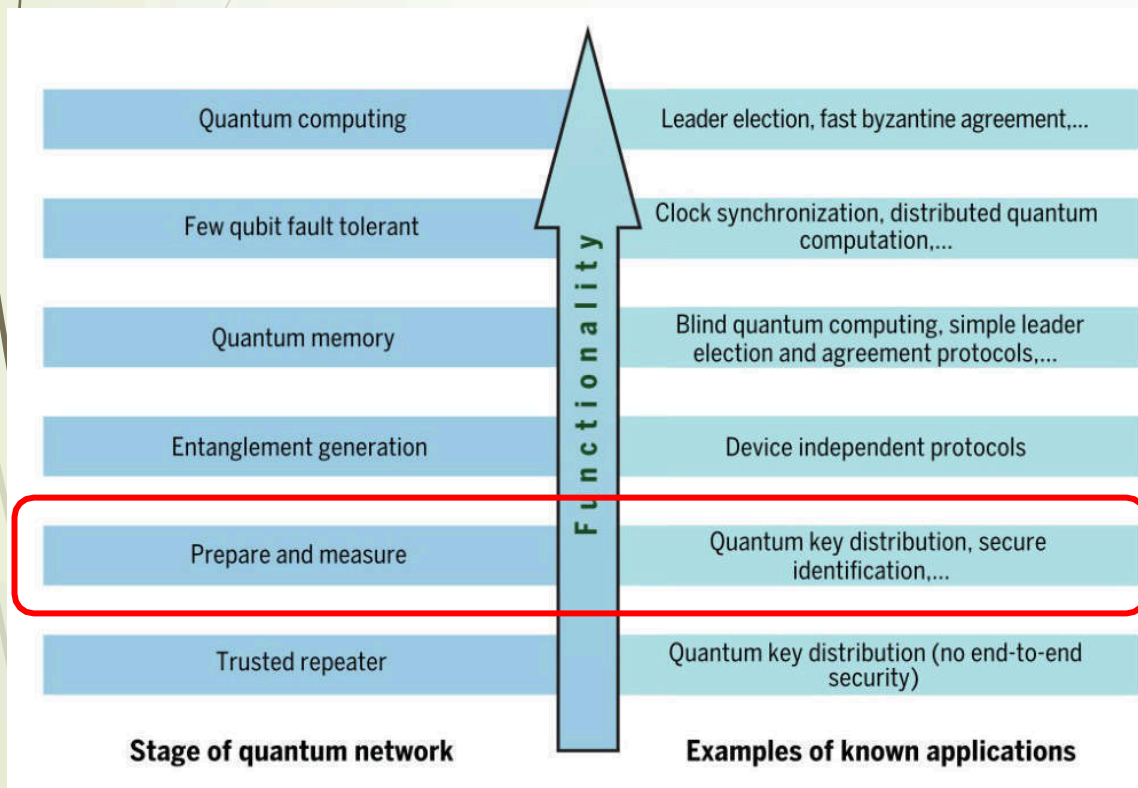


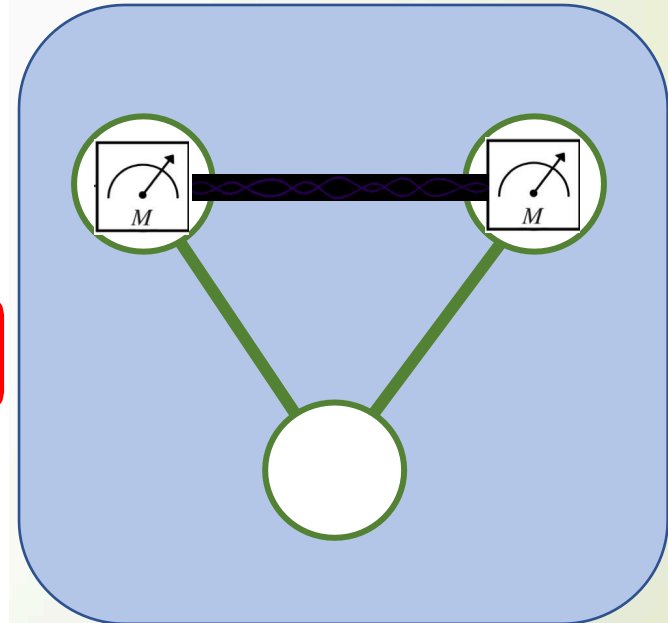
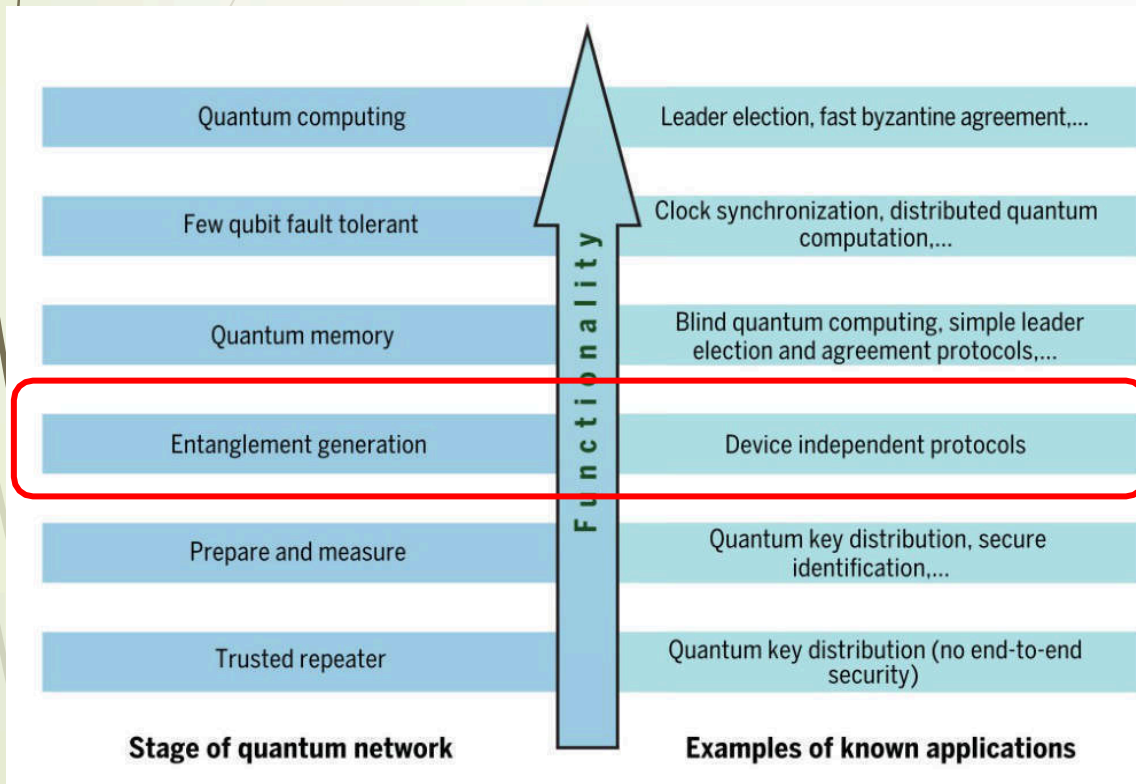
Quantum Key Distribution



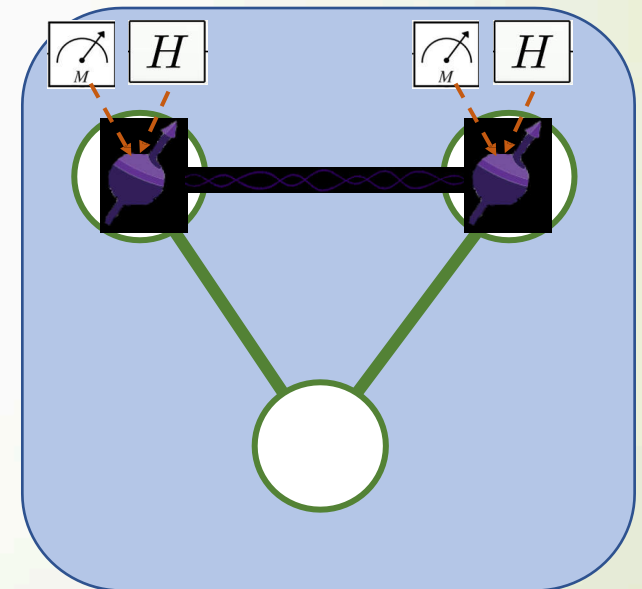
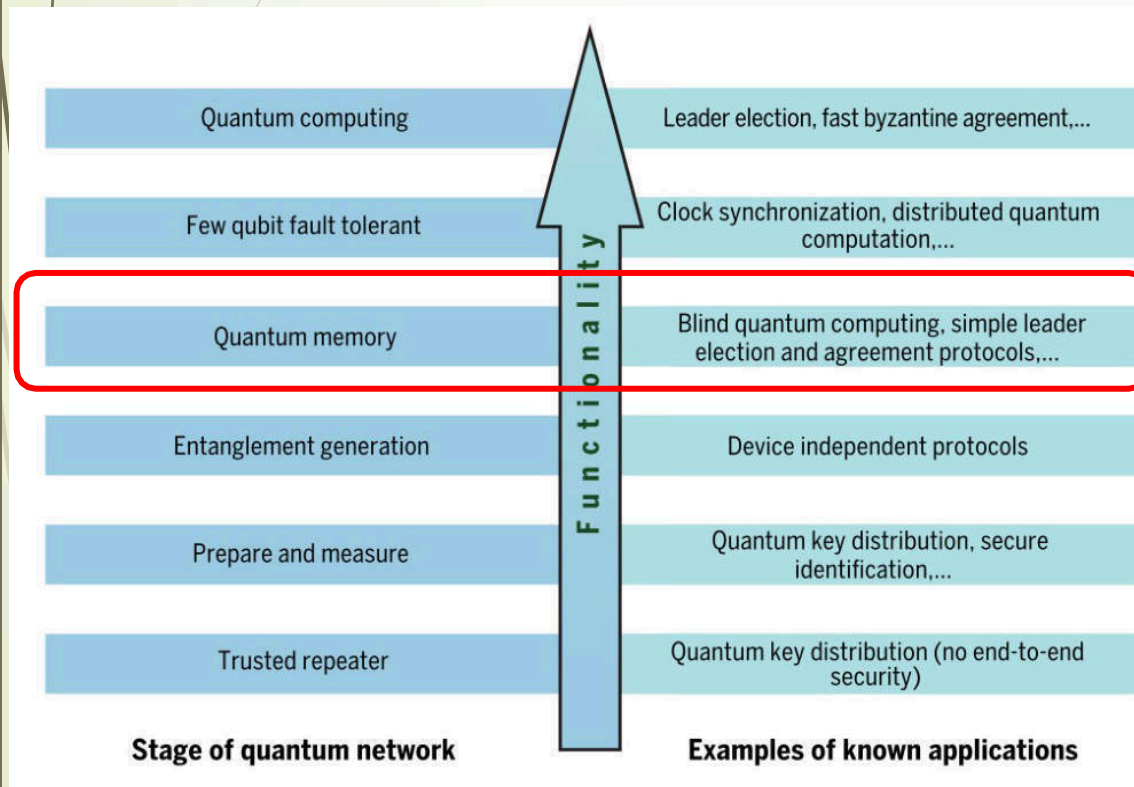
Secure Identification



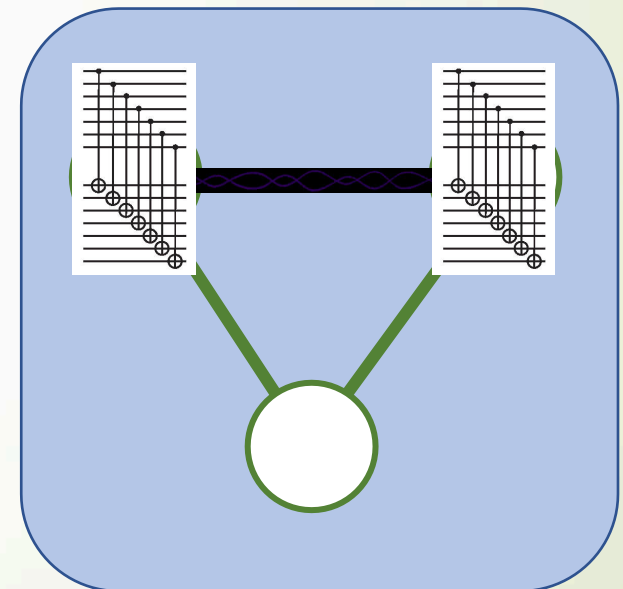
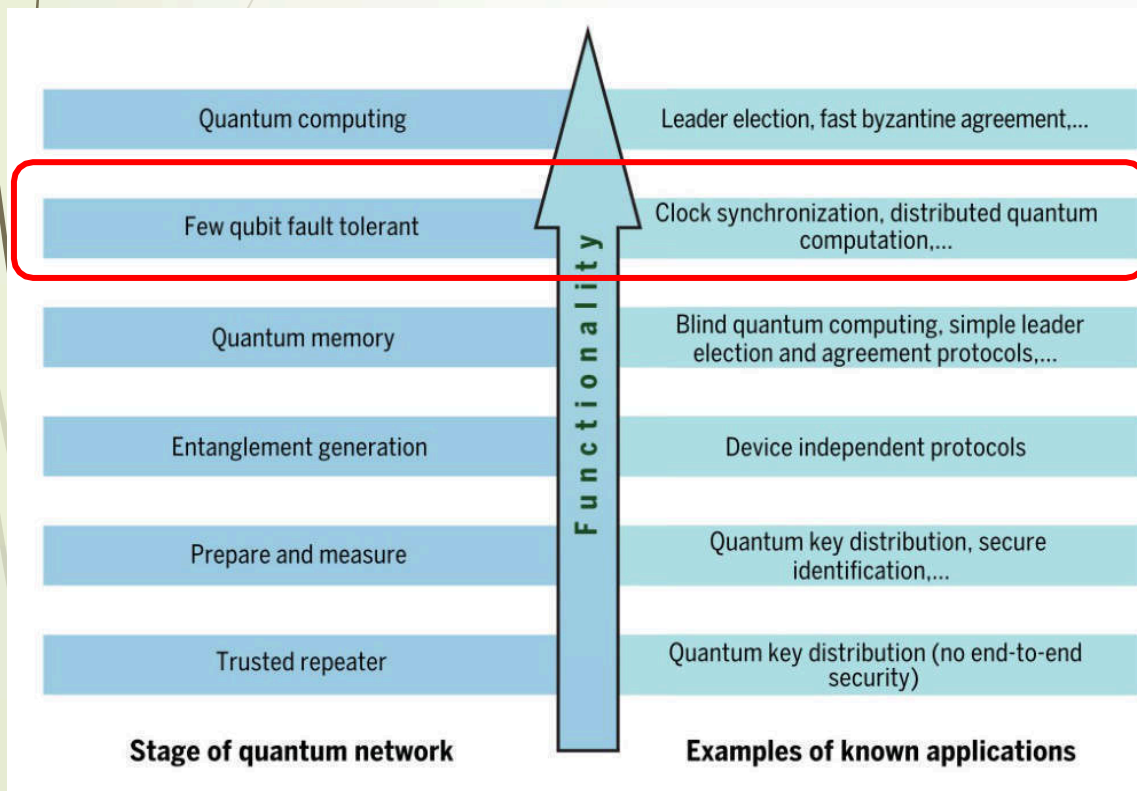
Device Independent Protocols



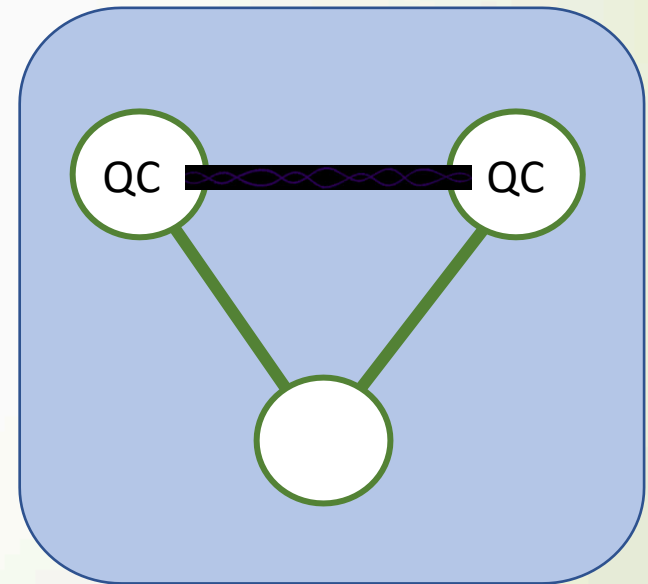
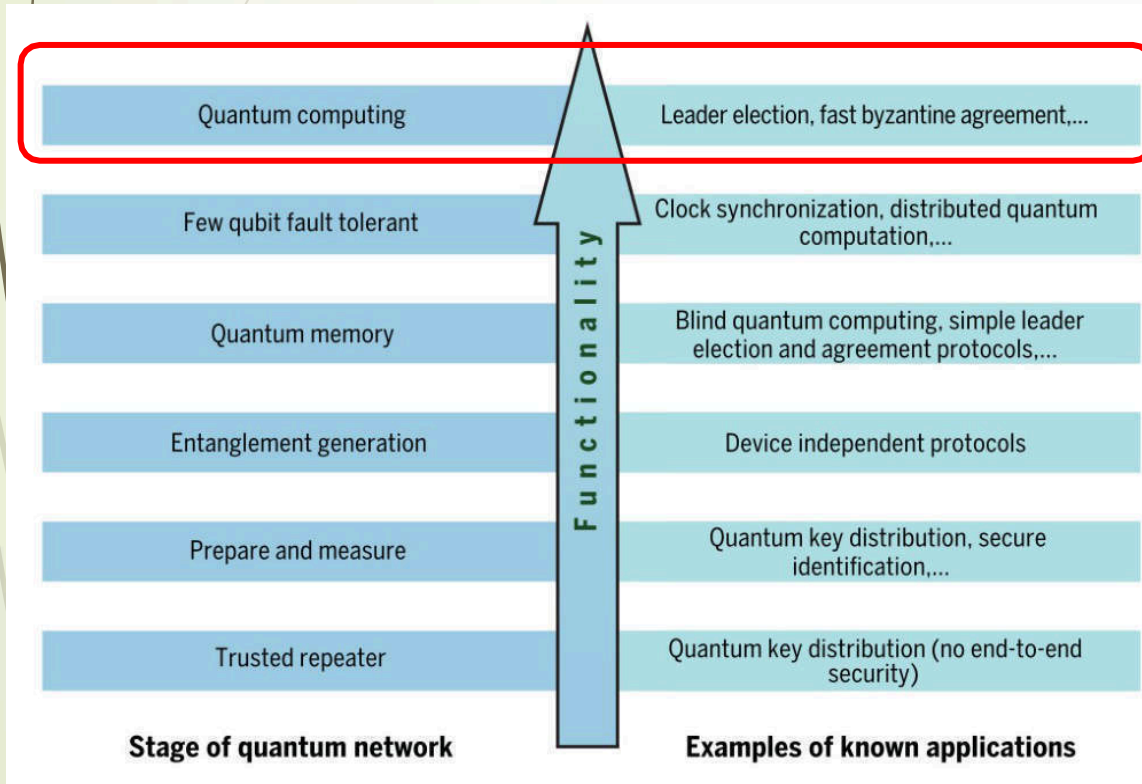
Blind Quantum Computing



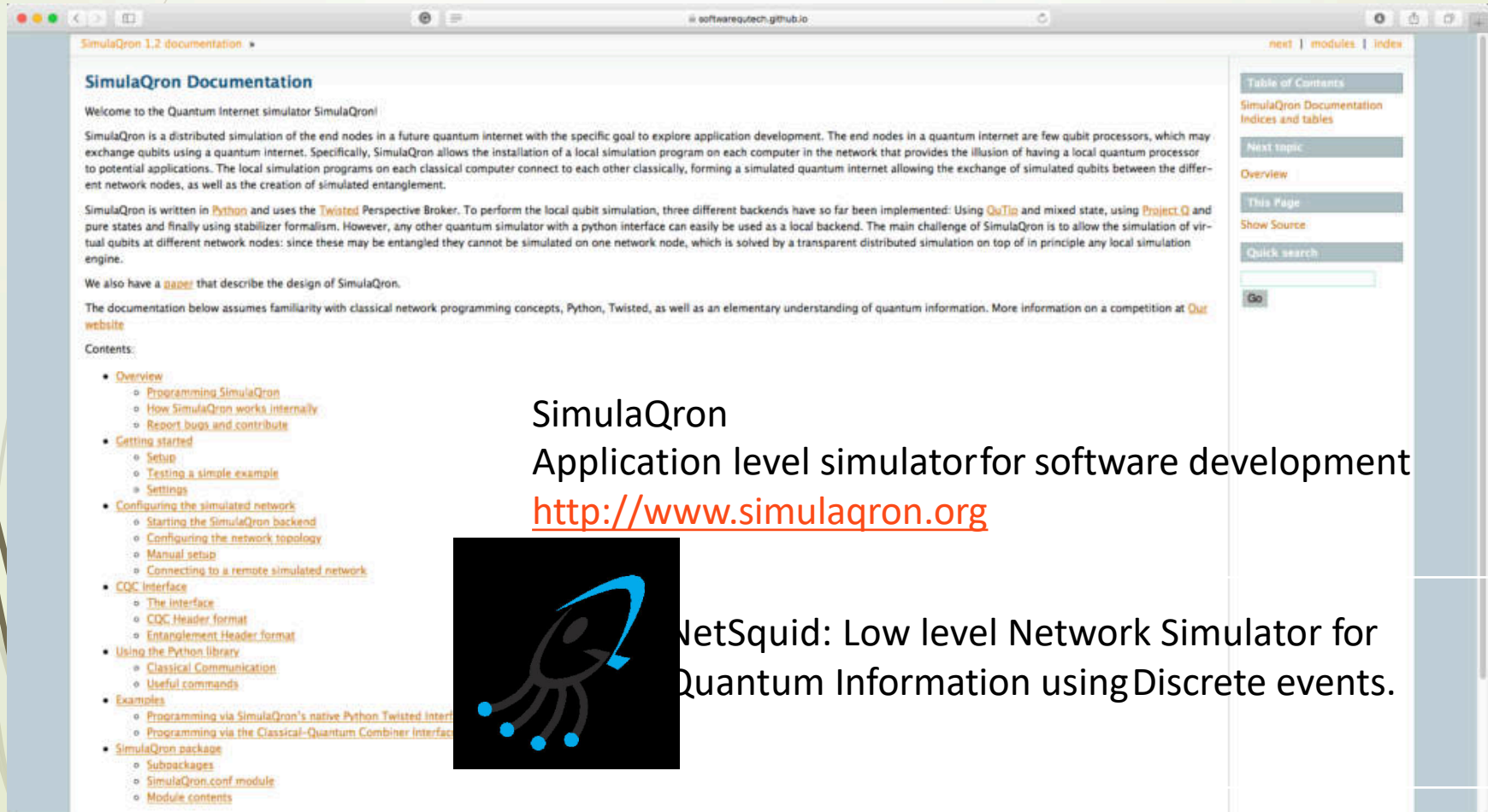
Clock Synchronization, Distributed Quantum Computing



General Distributed Quantum Computing



Quantum Internet Simulators



SimulaQron Documentation

Welcome to the Quantum Internet simulator SimulaQron!

SimulaQron is a distributed simulation of the end nodes in a future quantum internet with the specific goal to explore application development. The end nodes in a quantum internet are few qubit processors, which may exchange qubits using a quantum internet. Specifically, SimulaQron allows the installation of a local simulation program on each computer in the network that provides the illusion of having a local quantum processor to potential applications. The local simulation programs on each classical computer connect to each other classically, forming a simulated quantum internet allowing the exchange of simulated qubits between the different network nodes, as well as the creation of simulated entanglement.

SimulaQron is written in [Python](#) and uses the [Twisted](#) Perspective Broker. To perform the local qubit simulation, three different backends have so far been implemented: Using [Qutip](#) and mixed state, using [Project Q](#) and pure states and finally using stabilizer formalism. However, any other quantum simulator with a python interface can easily be used as a local backend. The main challenge of SimulaQron is to allow the simulation of virtual qubits at different network nodes: since these may be entangled they cannot be simulated on one network node, which is solved by a transparent distributed simulation on top of in principle any local simulation engine.

We also have a [paper](#) that describe the design of SimulaQron.

The documentation below assumes familiarity with classical network programming concepts, Python, Twisted, as well as an elementary understanding of quantum information. More information on a competition at [Our website](#)

Contents:

- [Overview](#)
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 - [How SimulaQron works internally](#)
 - [Report bugs and contribute](#)
- [Getting started](#)
 - [Setup](#)
 - [Testing a simple example](#)
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- [CQC Interface](#)
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- [Using the Python library](#)
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 - [Programming via SimulaQron's native Python Twisted interface](#)
 - [Programming via the Classical-Quantum Combiner interface](#)
- [SimulaQron package](#)
 - [Subpackages](#)
 - [SimulaQron.conf module](#)
 - [Module contents](#)

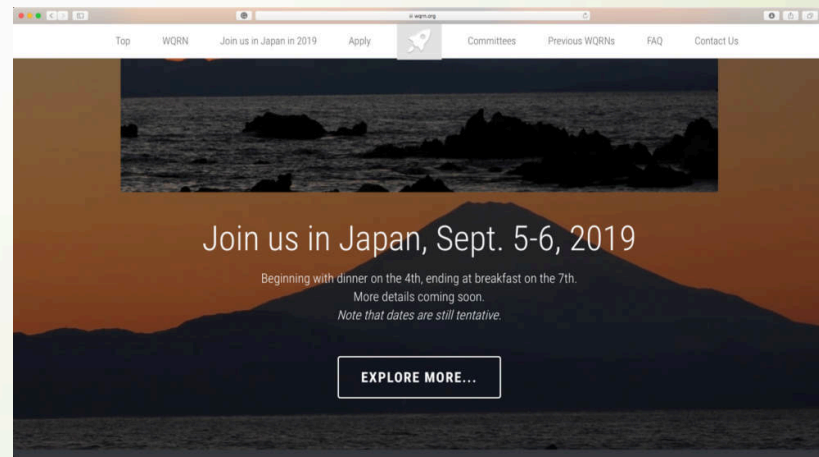
SimulaQron
Application level simulator for software development
<http://www.simulaqron.org>



NetSquid: Low level Network Simulator for Quantum Information using Discrete events.

Quantum Internet Research Group (QIRG)

- Idea originated in January 2018
- Presented at IRTF meeting in March 2018
- (Open) mailing list started from March 2018
- Approved as an RG on 2018/11/6
- Meet 3 times a year
 - 1x/year @IRTF (Prague)
 - 1x/year @quantum conference (<https://www.wqrn.org>)
 - 1x/year virtual/online



Questions

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